

## Synergistic Activities

### Gary L. Welz

**1. AI-Powered Research Tools for Scientific Synthesis** Developer and principal architect of the CopernicusAI Knowledge Engine (2023–present), an open-source platform integrating multiple AI models with academic databases to generate evidence-based research briefings with verifiable citations. All components publicly available on Hugging Face and GitHub under MIT License.

**2. Process Visualization as Computable Research Artifacts** Created the Programming Framework methodology for converting textual process descriptions into structured, computable diagrams using LLM-powered extraction and Mermaid visualization. Applied to biological sciences via Genome Logic Modeling Project (GLMP) with biological processes mapped as interactive flowcharts and JSON storage.

**3. Internet-Based Scientific Publishing and Multimedia Distribution** Early innovator in scientific communication via internet technologies (1989–2002). Founded Science Television Company (multimedia lectures distributed by American Mathematical Society). PI for Science and Engineering Television Network Project (\$60,000 Sloan Foundation grant, 1991–1992). Published on internet video distribution in ACM Interactions (1994) and international conferences.

**4. Mathematics Education Technology Integration** Implemented web-based mathematics learning systems across CUNY (2003–2020): Maple, Mathematica, Excel, ALEKS, MyMathLab, WebAssign, Khan Academy. Rubin Museum Fellow (2013): integrated Mathematics and English composition for first-year students.

**5. Open Science and Accessibility Advocacy** Public release of all research tools and datasets under open-source licenses. CopernicusAI components designed for WCAG 2.1 AA accessibility. Prototypes on Hugging Face Spaces for community access and validation. Provenance tracking and multi-checker validation for responsible AI.